

Landscape of Text Classification with Machine Learning in Official Statistics: Experiences from the AIML4OS Project

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Abstract

Recent years have witnessed remarkable progress in Natural Language Processing (NLP), particularly with the Transformer revolution and the subsequent emergence of generative models. These advances hold significant implications for National Statistical Institutes (NSIs), which routinely handle large volumes of text requiring classification according to major international and national nomenclatures such as ISCO (International Standard Classification of Occupations), NACE (Statistical Classification of Economic Activities), and COICOP (Classification of Individual Consumption According to Purpose), as well as their national adaptations and other less widely known statistical classifications.

Integrating this technological evolution into statistical processes aligns directly with NSIs' core mission towards innovation, as outlined in the Quality Assurance Framework of the European Statistical System: "The overall methodological framework used for European Statistics follows European and other international standards, guidelines, and good practices, while constantly striving for innovation." NSIs are thus expected to stay abreast of methodological advances that can enhance statistical processes—particularly in areas where the integration of Machine Learning (ML) and Artificial Intelligence (AI) into text classification can lead to notable improvements in accuracy (Principle 12: Accuracy and Reliability), timeliness and punctuality (Principle 13), and cost-effectiveness (Principle 10).

This drive for innovation is at the heart of the European project AIML4OS¹, and especially of its dedicated work package "From text to code - Experiences and potential of the use of AI/ML for classifying and coding" (WP10)². Eight European countries actively participate in this work package, with five additional countries contributing as observers. For these NSIs, applying ML/AI to text classification has become increasingly critical in recent years. WP10 leveraged the unique opportunity of having many NSIs aligned on the same subject to compile an overview of current developments and assess the state of ML/AI-based text classification within official statistics. Through presentations from participating NSIs³, WP10 collected an overview of current practices, challenges, and priorities in ML/AI-based text classification. This paper summarizes these contributions and presents the current landscape of ML/AI adoption

¹ <https://cros.ec.europa.eu/dashboard/aiml4os>, <https://github.com/AIML4OS>

² <https://cros.ec.europa.eu/book-page/aiml4os-wp10-text-code-experiences-and-potential-use-aiml-classifying-and-coding>, <https://github.com/AIML4OS/WP10>, <https://aiml4os.github.io/WP10/>

³ <https://github.com/AIML4OS/WP10/tree/main/ProjectsOverview/ProjectsPresentations>

for text classification in official statistics, highlighting common challenges and priority areas for future work.

Keywords: statistical classification, natural language processing, automatic coding

1. Introduction

The synthesis of experiences from participating and observing countries in WP10⁴ shows that the classification of textual data according to large and complex international statistical nomenclatures—such as ISCO⁵, NACE⁶, and COICOP⁷—represents a prominent use case for ML/AI-based text classification in official statistics⁸.

These international standards form the backbone of European statistics, ensuring harmonization, comparability, and coherence across member states and over time⁹. They provide a common framework for describing economic activities, occupations, and consumption, both across countries and over time. However, applying these nomenclatures in practice is a highly complex task. The classifications involve hundreds or even thousands of fine-grained categories. Choosing the appropriate class often requires technical expertise, a deep understanding of the specific context, and the ability to distinguish between nuanced cases. Moreover, many NSIs must deal not only with the European standard versions of these nomenclatures, but also with national adaptations, adding another layer of complexity. Historically, this task was performed manually by domain experts who had a strong command of the relevant nomenclature. In many cases, these experts would manually classify each text entry, sometimes even contacting respondents directly to verify unclear or ambiguous information.

This manual process, while focusing on precision, is time-consuming, resource-intensive and can also encounter classification errors. It also suffers from limited transparency and reproducibility: expert decisions often rely on implicit knowledge and cannot always be formalized into explicit, generalizable rules.

In recent years, however, significant progress in Natural Language Processing has opened the door for automating these classification tasks using Machine Learning and AI methods. As a result, many NSIs have begun experimenting with, and in some cases implementing, ML-based approaches to support or fully automate text classification. These efforts aim to increase efficiency, consistency, and reduce reliance on scarce expert resources—while maintaining or even improving classification quality.

2 Availability and Challenges of Training Data

2.1 Access to Labeled Historical Data

However, the successful application of ML methods largely depends on the availability and quality of training data. In many cases, NSIs benefit from access to large historical datasets manually labeled by domain experts. With deep knowledge of the nomenclature, these experts generally classified records carefully, though time pressure sometimes reduced accuracy. As

⁴ Destatis (Germany), Statistics Poland, Statistics Norway, Statistics Netherlands, Statistics Austria, Insee (France), National Institute of Statistics (Spain), Statistics Denmark, SURS (Slovenia), CSO (Ireland) and ISTAT (Italy) presented their use-cases within WP10: <https://github.com/AIML4OS/WP10/tree/main/ProjectsOverview/ProjectsPresentations>.

⁵ International Standard Classification of Occupations

⁶ Statistical Classification of Economic Activities in the European Community

⁷ Classification of Individual Consumption According to Purpose

⁸ Refer to the Appendix 6.1 “Projects Overview through WP10 presentations”

⁹ In line with the European Statistical System Quality Assurance Framework (QAF), notably indicators 7.2 and 7.4 on methodological consistency and comparability, as well as indicator 14.3 on coherence and standardization practices.

a result, many countries are able to train models using substantial volumes of high-quality labeled data.

Insee and Destatis, for instance, have leveraged extensive resources: the French NACE use case relies on c. 3 million labeled records, while German COICOP and NACE projects are built on datasets of about one million and four million entries, respectively. Statistics Norway, though working with smaller datasets, has successfully used 22,000 manually coded entries from a pilot Time Use Survey and 60,000 labeled items from the Household Budget Survey (HBS) for COICOP classification. At Statistics Denmark, three waves of data about three groups of products had been classified over the years, amounting to about 50,000 records in total.

This favorable situation is not universal. Some NSIs face resource constraints that limit their ability to create new labeled datasets. At SURS (Slovenia), the limited availability of expert manpower has led to the reuse of previously classified data coming from a different source. In their scanner data project on COCOIP classification, they reused already labelled product data from a previous retailer collaboration and applied synthetic perturbations to simulate OCR errors and boost model robustness. Statistics Netherlands chose to label a particular subset of 5-digit NACE codes in Section R on tourism and recreation (16,050 manually checked units) for an in-depth study on the performance of different feature sets for NACE prediction.

When human resources for labelling are scarce, some NSIs combine multiple labeled data sources originating from different groups of annotators—often with varying familiarity with classification schemes. Statistics Spain, for example, assembled around 800,000 descriptions labeled with the national NACE version from several sources. These records were annotated either by survey respondents, who typically possess limited knowledge of NACE, or by Statistics Spain employees. Even among the latter, the labeled data came from multiple teams within the institute. While all contributors can be considered experts in their respective domains, their familiarity with the NACE classification and the specific data varied significantly, leading to heterogeneity in labelling quality.

2.2 Low-Quality and Ambiguous Input Texts

Even when large training datasets are available, they are often noisy, inconsistent, or require significant pre-processing. Most projects deal with brief and poorly structured texts — such as product names from receipts, web-scraped content, short free-text survey responses, or job descriptions. These inputs tend to be ungrammatical, contain spelling mistakes (originating from typos or OCR errors), and show substantial lexical variability. Textual data also suffer from semantic distortions: ambiguity, where a word's meaning depends on context; homonymy, where the same form refers to different concepts; and synonymy, where several terms refer to the same concept. Such issues complicate the interpretation of meaning and contribute to inconsistencies in automatic or manual classification. An additional challenge arises when the text provided by households or companies is too general or lacks sufficient detail, making it very difficult to determine the precise category, even for human experts. Finally, the classification systems themselves can introduce further complexity, as some categories are very close in meaning or overlapping in scope. In such cases, even a human expert may hesitate between two or more classes that appear equally plausible, leading to potential inconsistencies or subjectivity in the final coding.

2.3 Inconsistencies and Labelling Uncertainty

As a consequence, these difficulties with textual data result in inconsistencies and inaccuracies in the labels used to train Machine Learning models, issues that are particularly pronounced under constraints of limited human resources and time pressure. Consequently,

many of these datasets may contain classification errors or inconsistencies that are difficult to detect or quantify. Properly assessing this would require re-annotating the same records by multiple experts to measure agreement—an approach that would significantly increase the manual workload, which is precisely what ML aims to reduce.

In the French NACE use case, for example, a large annotation campaign was launched to support a nomenclature revision. As part of this effort, a quality check was performed on a small subsample of the data by having each record independently labeled by two different experts. The resulting agreement rate was around 70%. A similar issue was observed in the Norwegian ISCO/NACE use case. A prior study based on SILC data examined the consistency of manual occupation coding by comparing classifications done by different coders. The observed disagreement ranged from 7% to 18% at the 2-digit level, often due to vague or ambiguous job descriptions.

While this level of agreement is still relatively high, it also highlights the intrinsic subjectivity of the task—and many other projects lack the resources to even assess this kind of classification variability.

To improve data quality, some NSIs have initiated manual reviews of their historical datasets. In the Spanish NACE use case, for example, a sample of records from different databases is manually re-labeled and checked to assess and improve classification quality. Both the text descriptions and the assigned codes are reviewed to ensure accuracy. The European Web Intelligence Network (WIN) project¹⁰ on using web data for business register enhancement listed four options to cope with that: 1) Manual editing: creating a "gold set", 2) Record selection: selecting codes that have a lower probability of being incorrect, 3) Use survey data and/or 4) Accept the errors and use a ML model robust for those errors.

Another layer of complexity comes from the heterogeneity of data sources. In the Spanish NACE project, training data is drawn from a wide range of business statistics using CNAE-2009 codes—both short-term and structural (such as the Structural Business Survey and the Business Register)—as well as from more recent ad hoc surveys based on either CNAE-2009 or the new CNAE-2025. These sources differ not only in time period and purpose, but also in classification version, making integration and harmonization of the training dataset particularly challenging.

2.4 Complexity of the Outcome Variable

Finally, the nature of the classification target itself adds to the challenge. International nomenclatures such as NACE, COICOP, or ISCO involve hundreds or thousands of hierarchical classes. Training data is typically sparse at the most detailed levels, and class imbalance—reflecting real-world distributions—can hinder ML performance. Underrepresented classes are especially difficult to model effectively. The outcome variable is often a national adaptation of international standards, typically extended with additional, more detailed levels, which can increase the complexity.

To address this, several countries are experimenting with strategies such as synthetic data generation, hierarchical classification schemes, or threshold-based filtering.

¹⁰ https://github.com/WebIntelligenceNetwork/Deliverables/blob/main/WP3/D3_11_WP3_UC5.pdf

3. Methodological choices

3.1 Synthetic Strategies

Some countries, particularly INE and Destatis, invest in generating synthetic training data from titles and explanatory notes of the classification system and enriching this dataset through a dictionary of synonyms. This approach is valuable for a wide range of scenarios: it can help to enrich quite complete training sets, as well as to build training sets for when scarce or no real data is available.

3.2 Similar Feature Engineering and Embedding Strategy

Most NSIs apply fairly standard text pre-processing techniques as a first step: removing stopwords, rare terms, punctuation marks, special characters, and numbers; converting all text to lowercase; and occasionally performing stemming or lemmatization.

An interesting and recurring observation across several projects is the effectiveness of character-level tokenization combined with n-gram features, particularly when working with very short text inputs. This strategy was notably successful in the German COICOP use case, where a significant performance improvement was observed after switching to character-level tokenization with n-grams. A similar result was reported by Statistics Norway in their COICOP project, where extensive testing showed that 2- and 3-character n-grams, combined with count vectorization, outperformed other methods. Statistics Denmark experimented with this approach as well but did not find character n-grams to be crucial.

This approach aligns closely with the strategy adopted in the French and Spanish NACE use cases, where fastText was used. FastText operates on a bag-of-character-n-grams basis, effectively capturing subword information and making it particularly suitable for sparse or noisy texts.

Statistics Netherlands experimented with different feature sets, such as, “yet another keyword extractor” (YAKE), so called “descriptive words”¹¹ and “concept words”¹², “improved global feature selection scheme” (IGFSS), the same with additional cleaning (IGFSS+) and combinations of these. They found that a combination performed best.

While many countries have expressed interest in exploring more modern embedding methods—particularly those based on Transformers—their implementation remains limited in practice, mainly due to infrastructure constraints, such as the lack of GPU access in several NSIs.

Statistics Norway provides an illustrative exception in their ISCO use-case, where they tested BERT-based models trained on Norwegian text, including versions such as NorBERT developed by the University of Oslo, and further fine-tuned on occupation texts. Since transformer embeddings result in high-dimensional dense vectors, Principal Component Analysis (PCA) was used to reduce dimensionality before feeding them into classification models. The results, especially with NorBERT, were promising—and importantly, the models were optimized to run on CPU-only environments, making them more feasible for deployment in NSIs with limited computing resources. At Statistics Austria or Statistics Portugal, meanwhile, projects have successfully trained Transformer models from scratch for use as text classifiers. Statistics Denmark fine-tuned a BERT model on their Danish scanner data. To have

¹¹ Knowledge-based words specific to NACE codes that manual editors currently use at CBS when they check business’ activity descriptions

¹² Abstractions of D-words. For instance the D-words station wagon, vehicle, four-wheel drive are examples of the C-word car. First D-words are selected in the text and next they are replaced by their corresponding C-words. In the past, CBS made knowledge graph that contained both C- and D-words for each of the NACE classes.

enough training data, they simplified the task to just a few broad categories (e.g., 4 out of 7 levels in the PPP¹³) and trained for hundreds of epochs, which risked overfitting and poor performance on unseen product descriptions. Statistics Spain has also applied a Spanish pre-trained BERT-based model (BETO) for coding the national version of NACE. Due to hardware constraints, it was trained with a subset of the complete training set fed to fastText (the size was reduced to 3% of the original set), producing slightly worse performance, thus showing the potential of BERT-based compared to fastText.

Still, these approaches remain a step behind the current state-of-the-art in embedding techniques, which rely on modern pre-trained large language models (LLMs) offering richer semantic representations.

In this regard, Istat (Italian National Institute of Statistics), within the PRODCOM project, used OpenAI's text-embedding-3-large model to generate embeddings from concatenated product names and descriptions. These texts were retrieved from company websites and preprocessed using the GPT-4o mini model to rephrase and simplify the original content. The experiments were conducted on OpenAI model instances deployed in ISTAT's private cloud environment.

Exploring the potential of such advanced methods—particularly for NSIs with the necessary infrastructure—is one of the core objectives of WP10. A key focus is to assess how these cutting-edge models can be realistically integrated into official statistical workflows.

3.3 Lightweight Models Favored for Production

Lightweight models such as fastText, logistic regression, and tree-based classifiers continue to be widely adopted by national statistical institutes (NSIs) due to their transparency, rapid training times, and minimal computational demands. These characteristics are particularly valuable in environments where legal or infrastructural limitations restrict the use of cloud services or prohibit the deployment of more resource-intensive solutions, such as pre-trained Transformer models. These approaches do not account for the hierarchical structure of the nomenclature. Nonetheless, experiences from several countries including a review of the literature by Statistics Austria and Statistics Denmark within WP10 suggested that hierarchical structures did not clearly and substantially increase the leverage over the classification problem¹⁴.

In recent years, fastText has attracted renewed interest, particularly among NSIs already operating models in production or preparing for deployment. It forms the core of the ML pipelines at both INE and Insee, where institutions have highlighted the strong performance achieved with this approach—often sufficient for many classification tasks in official statistics. However, fastText is now considered outdated. The official library has been archived by Meta, raising several concerns. The methodology, while efficient, no longer reflects recent advances in NLP, and the lack of ongoing maintenance introduces medium-term risks—such as the absence of bug fixes and growing incompatibility with modern software ecosystems. These issues may ultimately affect the reliability and maintainability of production systems built around fastText.

As a result, some NSIs are already exploring alternatives. For example, Insee has developed a custom framework called torchFastText¹⁵, built on PyTorch. While its current architecture mirrors that of fastText, it provides a flexible foundation for future developments.

¹³ *Purchasing Power Parities*

¹⁴ <https://github.com/AIML4OS/WP10/tree/main/LiteratureReviews>

¹⁵ <https://github.com/InseeFrLab/torch-fastText>

The framework allows for the integration of more advanced neural architectures, the use of Hugging Face models, and the implementation of uncertainty-aware methods such as explainability tools, model calibration, and conformal prediction.

Istat has developed and is currently testing two lightweight models for the classification of company products within the PRODCOM project. The first model is a large language model obtained through fine-tuning of LLaMA-3.2-3B-Instruct, while the second is a deep learning model. In both cases, training was performed using synthetic data derived from a mapping table that links PRODCOM classification codes to the corresponding descriptions in NC8 (Combined Nomenclature). This approach was adopted because NC8 provides a more detailed set of product definitions than the standard PRODCOM classification system.

3.4 Aligned Evaluation Framework

Hyperparameter tuning through cross-validation was a critical step in every project. Whether using simple train/test splits or more robust k-fold cross-validation, all teams emphasized the importance of rigorous evaluation to ensure generalizability and model stability.

Given the often severe class imbalance in classification tasks based on statistical nomenclatures, most NSIs focused not only on overall accuracy but also on more nuanced metrics such as precision, recall, and the F1 score. These indicators were especially important for rare classes, where errors can have disproportionate effects on statistical outputs. Several projects also stressed the importance of evaluating model performance not only at the top-level class hierarchy but also at the level of sub-classes, where classification becomes more ambiguous and error-prone.

Equally crucial was the assessment of confidence scores—that is, the model’s estimate of how likely a given prediction is to be correct. These scores are often derived from the predicted class probabilities and provide essential information for implementing human-in-the-loop strategies. Rather than relying solely on aggregate metrics like accuracy or F1 score, many teams paid close attention to how well the model could self-assess its predictions.

The Norwegian COICOP use case illustrates this well. In their evaluation, two models showed similar overall accuracy. However, when focusing on high-confidence predictions—those that could be classified automatically with minimal risk—the differences were striking. The final setup allowed 37% of items to be automatically classified at an estimated 90% accuracy level, compared to only 14% in the earlier system. This highlighted that confidence calibration is just as important as overall accuracy when the goal is to reduce manual effort without sacrificing quality.

4. How ML is used within the Classification Task

4.1 Different Levels of Maturity Across NSIs

Although text classification using ML/AI is a common interest among all WP10 participants, countries are not all at the same stage of development.

Some, like Insee, Statistics Norway, Statistics Austria or Statistics Poland, have already been using ML models in production for several years. Others, like Destatis, have completed a first production run for the Household Budget survey but with limited automation—e.g., using a model stored on a standalone server, disconnected from the main survey management tools, thus preventing end-to-end dataflow automation. INE and CSO are currently transitioning to production, while NSIs like SURS are still in the pilot phase. Some, including Statistics Denmark, are only beginning to explore the potential.

The landscape is therefore marked by considerable variation in maturity and implementation strategies across NSIs.

4.2 Varied Roles of ML in the Classification Pipeline

Even among countries with production systems in place, the integration of machine learning into the classification process varies significantly.

In some implementations, such as the German and Polish COICOP projects, the French NACE project, and several Norwegian use cases, machine learning models are used for direct classification—assigning a single class automatically to each record with a confidence score above a predetermined threshold, without human intervention, at least for those cases.

By contrast, other national statistical institutes employ machine learning primarily as an aid to human coders, generating a shortlist of likely classes to assist in manual decision-making. This assistant role is exemplified by Ireland’s Central Statistics Office, where coders retain full control and make the final determination based on the model’s suggestions. Statistics Austria follows a similar approach, leveraging machine learning to produce the top k predictions, which are then carefully reviewed by human annotators. INE employs a hybrid strategy, leveraging automatic classification when the model’s confidence is high and routing uncertain cases to coders, who then work from a list of recommended codes. Statistics Netherlands takes yet another approach with a subproject focusing explicitly on detecting misclassifications in the business register. The variety in the use of machine learning methods in the statistical process pipeline reflects differing operational needs, implementation choices, and levels of acceptable automation within a shared quality assurance framework.

Machine Learning is not always the first method applied in the classification workflow. In several countries, a preliminary matching step is used to identify whether the input text has already been classified in the past, using algorithms based on string similarity measures—ranging from exact string matching to fuzzy methods like Levenshtein distance or Ratcliff-Obershelp. Only when no match is found does the Machine Learning model take over. In Norway, this approach is applied systematically across projects and Poland follows a similar approach in its COICOP pipeline.

These approaches should not be interpreted as institution-wide strategies, however, as many national statistical institutes (NSIs) adopt different levels of automation and human involvement depending on the statistical domain, operational requirements, data quality constraints, and maturity of the production process.

4.3 Human-in-the-Loop Design and Confidence Scoring

Despite advances in model accuracy, few systems operate fully autonomously. Instead, most pipelines include human-in-the-loop mechanisms to allow expert review, correction of borderline cases, and contribution to model improvement over time. Confidence scores are widely used to manage this process: predictions with low confidence are flagged for manual verification, while high-confidence classifications can be processed automatically.

However, the threshold for what is “high enough” varies across countries and depends on statistical requirements, institutional risk tolerance, and operational constraints such as available time and human resources. In the Polish COICOP project, for instance, all Machine Learning outputs are reviewed by experts to ensure that no errors affect published indicators. Other NSIs try to automate as much as possible, accepting a degree of residual error in exchange for significant efficiency gains.

Beyond the classification logic itself, the design of the human-in-the-loop interface plays a critical role. The way predictions are displayed—whether a single class or a ranked list—is not a trivial detail. Some systems also show the confidence score or allow coders to browse similar

cases. The structure and context in which items are presented also matter: grouping related entries or allowing contextual information from neighboring records can help coders make quicker and more accurate decisions. Getting this design right has proven essential for reducing cognitive load and improving the quality and speed of manual verification, particularly in high-volume settings.

4.4 Monitoring and Retrainings for Models used in Production

Retraining is essential for all countries operating classification projects in production, as it helps mitigate the risk of model drift. Since most countries implement human-in-the-loop workflows, newly labeled data from ongoing processes is regularly used to update and retrain models. The retraining frequency varies, but annual updates are common. However, comprehensive model monitoring strategies are rarely implemented. A notable exception is the French NACE use case, where samples are manually reviewed and compared against machine learning outputs via a dedicated dashboard. Statistics Austria is also making significant progress by developing a monitoring dashboard to track production model performance over time.

4.5 Legal and Data Protection Concerns

Data sensitivity and legal compliance are a consistent constraint. Most participants avoid using external APIs or cloud-based LLMs for sensitive content. Preference is given to open-source models, eventually trained or fine-tuned locally, and in some cases, only synthetic data or anonymized public sources are used.

5. Conclusion

Despite the varying local contexts, national statistical institute (NSI) projects across Europe are converging on a core set of methodological challenges, revealing significant opportunities for collaboration and knowledge sharing. This recognition prompted the creation of Work Package 10 (WP10) on text classification within the AIML4OS European project. WP10's aim is to systematically identify shared obstacles and to propose coordinated, cross-country actions, enabling NSIs facing similar issues to address them collectively. At the same time, the project fully acknowledges the heterogeneity of NSI environments, operational constraints, and institutional priorities. Rather than promoting a single standardized pipeline, WP10 seeks to document, compare, and disseminate a range of methodological solutions, allowing each NSI to adapt approaches to its own context and needs. To support transparency, reproducibility, and community-driven innovation, all project outputs — including literature reviews, code repositories, methodological reports, and tutorials — are being openly shared through GitHub. Through this collaborative and open approach, WP10 aims to contribute to the development of robust, scalable, and reusable solutions for text classification in official statistics. Beyond the immediate benefits for participating institutions, the project aspires to provide a long-term methodological resource for future NSIs facing comparable challenges in the integration of AI and machine learning into statistical production systems.

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6. Appendix

Projects Overview through WP10 presentations

Table 1: Projects Overview through WP10 presentations

Data Source	NSI(s)	Project Description
COICOP (Classification of individual consumption by purpose)		
Scraped data from pharmaceutical online shops	Statistics Poland	Classification of medical items for CPI calculation according to COICOP.
Retailer purchase data	Statistics Denmark	Classification according to COICOP for the calculation of Purchasing Power Parities (PPP) for a selection of product groups.
Scraped data from online computer shops	SURS (Slovenia)	Classification of computer-related products to build an average user's basket according to COICOP.
Scanned receipts from Household Budget Survey (HBS)	SURS (Slovenia)	Project 1: Classification of scanned receipts from Household Budget Survey (HBS) according to COICOP, using OCR-methods for text extraction. Training material comes from a different source, i.e. scanned articles from their biggest retailers. Focus on food, drink and tobacco articles. Project 2: Same training data was also used for a similar COICOP-project, classifying scanned articles already clean and labeled, with no actual OCR process involved.
Household Budget Survey (HBS)	Destatis (Germany)	Classification of household expenses according to COICOP.
User-entered or OCR-extracted purchase descriptions (HBS)	Statistics Norway	Classification of purchase data in HBS using short product descriptions according to COICOP.
Household Budget Survey (HBS)	Statistics Austria	Classification of household expenses according to COICOP.
Retailer purchase data	Statistics Portugal	Classification of web-scraped retailer purchase data according to COICOP.
NACE (Statistical classification of economic activities)		
Business Register (Self-	Insee (France)	Classification of company activities according to NACE classification.

declared company descriptions)			
Survey data, business register data and synthetic data	INE (Spain)	Classification of company activities according to NACE classification.	
Company descriptions from survey data	Destatis (Germany)	Classification of company activities according to NACE classification.	
SILC survey responses + register data	Statistics Norway	Classification of industries from living conditions survey and administrative registers.	
Labour Force Survey (LFS)	Statistics Austria	Classification of survey data on economic activities according to NACE	
Labour Force Survey (LFS), 2011 Census (Web responses)	Statistics Portugal	Classification of survey data on economic activities according to NACE	
Business Register and scraped website data	Statistics Netherlands	NACE prediction	
Business Register and scraped website data	Statistics Netherlands	NACE misclassification detection	
Business Register	Statistics Poland	NACE revision	
Business register, which includes descriptions from tax authority. Hopefully also scraped data from reports	CSO (Ireland)	Model which provides suggested NACE codes to users. Tool which incorporates this model, together with functionality for processing batches, choosing codes, and saving results. As well as using AI to provide suggestions, we also use tokenised string matching onto the classification labels, which is useful for rarer cases. This is enhanced by enriching the classification labels with LLMs.	
Web-scraped data on company products	ISTAT (Italy)	AI and ML-based automatic classification of company products scraped from corporate websites according to the PRODCOM nomenclature. PRODCOM ("PRODUCTION COMMunautaire" or Community Production) is derived from NACE and specifically covers the production of manufactured goods by enterprises in EU countries.	
Labelled company descriptions from CIRCE, ATECO 2022, and synthetic data. CIRCE is ISTAT's rule-based	ISTAT (Italy)	Evaluation of AI-based methods for automatically coding enterprise activity descriptions into the ATECO classification (Italian version of NACE). The project compares embedding-based approaches with traditional rule-based systems.	

tool for activity
coding support.

ISCO (International standard classification of occupations)

Labour Force Survey responses	SURS (Slovenia)	Classification of professions based on LFS answers.
SILC survey responses + register data	Statistics Norway	Classification of occupations using survey responses and employment registers.
Labour Force Survey (LFS)	Statistics Austria	Classification of survey responses on occupations according to ISCO.
Job Vacancy Survey (JVS)		
Structure of Earnings Survey (SES)		
Labour Force Survey (LFS) using training data from previous Census	Statistics Portugal	Classification of survey responses on occupations according to CPP, which is the set of all existing occupations in Portugal and their respective functional description, presented together by professional groups, but not exactly ISCO.

ACL (Activity Coding List)

Time Use Survey (TUS) responses	Statistics Norway	Classification of participants' daily activities using European ACL standard.
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ICD (International Classification of Diseases)

Death certificates	CSO (Ireland)	Classification of causes of death for mortality statistics. User tool for processing batches of certificates and viewing suggested codes. As well as using AI to provide suggestions, we also use tokenised string matching onto the classification labels, which is useful for rarer cases. We have a specialised spell-correction process which prioritises replacements which are from a medical dictionary, and which sound the same (soundex algorithm).
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ISCED (International Standard Classification of Education)

Labour Force Survey (LFS)	Statistics Austria	Classification of survey data on education according to ISCED.
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Other

Scraped job ads from company websites	SURS (Slovenia)	Identification and classification of online job advertisements using ML/NER to create alternative labour market statistics datasets.
Textual data & any classification scheme	Statistics Netherlands	Generic, semantically enriched classification with LLMs. Code name: DABING.
